

What the algorithm adds, in five everyday requests

BEFORE — WITHOUT COMPOSITION

Can't see which files are at risk, so it can only stop and ask — with nothing to restore if you say yes.

AFTER — THE FULL ALGORITHM

Names the exact files and backs them up first, so the deletion can be undone.

① “Clean up my old screenshots” `find . -name "Screenshot*.png" -delete`

BEFORE

```
command: find . -name "Screenshot*.png" -delete
composition: disabled (table alone, ablation)
class: unrecoverable
issue opaque: composition disabled (ablation): find not resolved through
```

AFTER

```
command: find . -name "Screenshot*.png" -delete
composition: enabled (full algorithm)
class: recoverable-with-capture
remove ./Screenshot 2026-09-20 184502.png
remove ./Screenshot 2026-09-14 101211.png
```

② “Delete the duplicate photos” `find . -name "* copy.jpg" -exec rm {} \;`

BEFORE

```
command: find . -name "* copy.jpg" -exec rm {} \;
composition: disabled (table alone, ablation)
class: unrecoverable
issue opaque: composition disabled (ablation): find not resolved through
```

AFTER

```
command: find . -name "* copy.jpg" -exec rm {} \;
composition: enabled (full algorithm)
class: recoverable-with-capture
remove ./sunset copy.jpg
remove ./beach copy.jpg
```

③ “Get rid of the zip files” `find . -name "*.zip" | xargs rm`

BEFORE

```
command: find . -name "*.zip" | xargs rm
composition: disabled (table alone, ablation)
class: unrecoverable
issue opaque: composition disabled (ablation): find not resolved through
issue opaque: composition disabled (ablation): xargs not resolved through
```

AFTER

```
command: find . -name "*.zip" | xargs rm
composition: enabled (full algorithm)
class: recoverable-with-capture
remove ./installer.zip
remove ./photos-backup.zip
```

④ “Delete my old drafts” `for f in *draft.docx; do rm "$f"; done`

BEFORE

```
command: for f in *draft.docx; do rm "$f"; done
composition: disabled (table alone, ablation)
class: unrecoverable
issue opaque: composition disabled (ablation): loop body not expanded
```

AFTER

```
command: for f in *draft.docx; do rm "$f"; done
composition: enabled (full algorithm)
class: recoverable-with-capture
remove ./resume draft.docx
remove ./thesis draft.docx
```

⑤ “Delete my OldPhotos folder in the background” `nohup rm -r OldPhotos &`

BEFORE

```
command: nohup rm -r OldPhotos &
composition: disabled (table alone, ablation)
class: unrecoverable
issue opaque: composition disabled (ablation): wrapper not resolved through
```

AFTER

```
command: nohup rm -r OldPhotos &
composition: enabled (full algorithm)
class: recoverable-with-capture
remove ./OldPhotos
```

What “composition” means: seeing through the parts of a command that hide its real targets — a search (`find`), a pipe (`|`), a loop, or a wrapper — to find the exact files it will touch. It is the only thing switched off in “Before”.

Across 777 test commands, switching composition off drops the share of lost data recovered from **89.5% to 35.4%** ($p = 2.1 \times 10^{-53}$). With it on, **no command destroyed data without asking** (0 of 405), and **9 in 10** commands that lose data were backed up and verified restorable; the rest are stopped for you to decide. Every screenshot above is the algorithm’s real, unedited output.